Brains share the rule, but not the wiring

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Abstract

Take a model that has watched several mice being perturbed, then show it a new mouse doing nothing in particular. From that ordinary activity and the settings of a perturbation about to be delivered, it has to say what happens next in an animal it has never seen perturbed.

In 20 mice whose frontal cortex was silenced with light, this works down to individual neurons: $\Delta R^2 = +0.134$ against $+0.039$ for the average of the other mice, a difference of $+0.095$ at $p = 3.5e-04$, reaching 0.6 in the best measured animals. The rule it exploits can be stated without a model. Across 38 mice the light suppresses most the cells that most sharply distinguish the two choices ($r = -0.171$, 32/38, $p = 2.4e-05$) and the cells that fire fastest ($r = -0.221$, 28/38, $p = 0.005$), while how much a cell ramps predicts nothing, a negative control inside the same analysis.

A score on the whole response is dominated by a component every neuron shares, so we split the effect into that component and the part specific to one neuron, on which a stereotype scores exactly zero. Individual structure transfers under light (15 of 20 mice, $p = 0.041$) and not under microstimulation of somatosensory cortex (3 of 6), where it is better resolved (a measurable maximum of $+0.895$ against $+0.574$). In the half of the light cohort whose individual response is best resolved, split on that maximum before any model is fitted, the operator recovers 22% of it in 9 of 10 animals ($p = 0.021$). The same measurement says why: under current, a neuron's ongoing activity says nothing about how much the stimulus will move it (-0.015, 4/6).

How much transfers grows with the size of the cohort the operator was fitted on and with how well each animal was measured, and fitting it on one animal is far worse than predicting nothing at all. So a shared causal rule does reach single cells, when the perturbation engages structure the animals have in common, and how far it reaches is set by what has been measured so far.

1 The question

Record the same brain area in two animals doing the same task and the population activity looks alike. Many methods now use that similarity, aligning recordings across sessions and subjects into a common space [1, 2, 5–7], and fitting latent dynamics that carry a behavioural decoder from one animal to another [3, 4].

All of that describes what activity looks like when the brain is left alone. It leaves open a question about mechanism. If two animals share a dynamical rule, then pushing them the same way should move them the same way. If the resemblance is only in the trajectories they happen to visit, then it need not.

We wanted a test that could tell those apart, so we set up a prediction problem. A model sees interventions in several animals. It then meets a new animal and is allowed to watch only its ordinary, unperturbed activity. Given the settings of a stimulus, it has to produce the whole time course of what follows, in the neurons and in the behaviour, before that stimulus is ever delivered to that animal.

Table 1: Every headline claim, the animals it is measured over, and the test. Animal-level throughout: an exact sign test over animals, which cannot be moved by one extreme value. The last column is the same analysis run where the answer is known to be zero.

claim	measure	animals	p	null
A shared operator predicts single-neuron responses to an unseen perturbation better than the average of the other animals	$+0.134$ vs $+0.039$	15/20	3.5e-04	–
How sharply a neuron distinguishes the two choices predicts how much the light suppresses it	$r = -0.171$	32/38	2.4e-05	-0.001, 18/38
How fast a neuron fires predicts how much the light suppresses it	$r = -0.221$	28/38	0.005	+0.000, 17/38
How much a neuron ramps predicts nothing (negative control)	$r = -0.003$	19/38	1.000	-0.041, 25/38
Under microstimulation, firing rate predicts nothing	$r = -0.015$	4/6	0.688	-0.131, 4/6
In the better measured half of the light cohort, split on a ceiling fixed before any model is fitted, the operator recovers a fifth of everything individual that is measurable	22% of ceiling	9/10	0.021	–
The individual part of the response transfers, under light	$+0.026$ median	15/20	0.041	-0.002
The individual part does not transfer, under current, though it is better resolved there	ceiling $+0.895$	3/6	1.000	-0.002
Fitting the operator on more animals keeps helping, with no sign of flattening	$r = +0.77$ with $\log n$	29/39	0.003	–

2 What we measured

We used two public datasets in which many mice received a perturbation described by numbers chosen by the experimenter, so that an unseen intervention has a meaning a model can be given.

Light in frontal cortex. 39 mice performed a delayed response task while anterior lateral motor cortex was silenced with light on a subset of trials [10–12]. The perturbation is described by its power, from 0.8 to 45 mW, and by where the light was aimed. We kept 189 sessions, 2,971 neurons, 10,391 light trials and 29,802 control trials, binned at 50 ms and aligned to the moment the light came on.

The cohort comes from three releases of the same experiment. The first describes the light with per trial columns; the other two do not, and instead carry the laser power as a continuous trace with onset and offset events, so a trial counts as perturbed when an onset falls inside it, its dose is the power recorded at that onset, and its site is whichever laser trace covers that moment. Sessions that silenced the pontine nuclei rather than cortex are dropped, because pooling them would pollute a stereotype built over dose. This is the cohort that carries the statistics: with 39 animals an exact test can reach $p = 3.6e - 12$.

Current in somatosensory cortex. 6 mice learned to report a stimulus delivered through a chronically implanted probe by turning a wheel. The stimulus is described by numbers we know in advance, which makes it easy to say what an unseen intervention means: the current, from 1 to 12 µA across 12 levels, and which contact on the probe delivered it, which sets its depth in cortex.

After quality control we kept 48 sessions, 949 sorted neurons, 16,532 stimulation trials and 13,044 windows of unperturbed activity, binned at 25 ms. Sessions record between 8 and 33 neurons at once, with a median of 19.

Three readouts run through the paper. The behavioural one is the probability that the animal has reported the stimulus by a given moment, which gives a smooth response curve that grows with current. The population one is the average firing rate across the recorded neurons. The finest one is the response of each neuron on its own.

We always score the effect of the stimulus rather than the raw activity, because a model can track activity well while knowing nothing about the intervention. Writing Δ for the measured change caused by the stimulus,

$$\Delta R^2 \;=\; 1 - \frac{\sum(\Delta - \widehat{\Delta})^2}{\sum \Delta^2},$$

so a score of zero is exactly the model that says the stimulus does nothing, and one is perfect. Because Δ is estimated from a finite number of trials, we also compute how high any model could possibly score, by splitting trials in half and comparing the two halves. That ceiling sits at 0.91 for neurons and 0.84 for behaviour, so the measurements themselves are not what limits us.

Nothing the model reads shares a trial with what it predicts. The effect is the perturbed mean minus the control mean, and the model is handed each neuron's control activity as an input. If the same trials produced both, their noise enters the input and the target with opposite signs, and a model can score above zero without knowing anything. So the control trials of every recording are split in two. One half builds every input the model sees. The other half defines the effect it is scored against. We found this mattered: before the split, the shared operator scored about twice as high on the quantity in Section 5, and half of that was noise. As a check, we reran the whole analysis with the perturbed trials replaced by a second group of control trials, so the answer is known to be zero, and it returns -0.002 (3 of 20 animals above zero).

Animals are the sample size. The claim here is about animals. Several sessions from one mouse are still one mouse, so every headline number is inferred at the animal level. Confidence intervals come from a bootstrap that resamples animals and carries all of their sessions along, and significance comes from an exact sign flip permutation over animals. With six animals the smallest p value that test can ever return is 0.031, reached when all six fall on the same side, and we say so wherever it appears; with 20 it is $1.9e-06$. Session level numbers appear alongside as secondary detail, and an earlier version of this work reported them as the headline, which overstated the evidence.

3 The model

The model is one operator shared by every animal, applied to whatever the new animal happens to be doing. For a neuron n in animal i we write the effect of the perturbation as

$$\widehat{\Delta}_n(t) \;=\; \underbrace{g(t,\theta)}_{\text{shared with everyone}} \;+\; \underbrace{\textstyle\sum_k m_{nk}(t)\, r_{nk}(t)}_{\text{acting on this neuron's own activity}} \;+\; a_n(t),$$

where θ collects the perturbation settings the experimenter chose, g is the average response of the other animals at that setting, and r_{nk} are channels measured from this neuron on control trials: its mean firing, how much that firing has grown since the alignment point, and the difference between the two trial types, which is what the neuron knows about the upcoming choice. The gains m and the additive term a come from a set transformer that attends across the neurons recorded together, so what a neuron is predicted to do can depend on the population it sits in. Every weight in it is shared across animals. Everything animal specific enters through r_{nk}, which costs nothing from the protocol because control trials are all we are allowed.

Writing the model as a correction to g matters more than the architecture does. Our first version had to learn the average response and the individual departures from it at the same time, and its errors on the easy part buried whatever it learned about the hard part. Starting from g, with the corrections initialised to zero, means every parameter is spent on the question we care about.

For a held out animal, g is built from the other animals and the network is trained on them, with one further animal set aside to decide when to stop. The held out animal contributes its control trials and the settings of a perturbation it has not yet received.

4 What transferred

4.1 Individual neurons, scored on the whole response

In the light cohort the model predicts individual neurons in a mouse it has never seen perturbed, and it beats the strongest simple alternative. The average of the other mice scores $+0.039$ $[+0.005, +0.080]$ over animals. The model scores $+0.134$ $[+0.065, +0.210]$, above zero in 15 of 20 mice, and the paired difference over animals is $+0.095$ at $p = 3.5e-04$ (Figs. 6 and 7). On the four best measured mice it reaches between 0.48 and 0.60 against a stereotype that manages between 0.08 and 0.41.

The network on its own scores $+0.106$, better on average than the stereotype but not reliably so ($p = 0.114$), because it fails badly on the animals whose response is barely resolvable. What closes that gap is deciding how far to trust it using two quantities that need no labels: how resolvable this animal's response is, which follows from trial counts and firing rates, and how much independently seeded copies of the network disagree with each other. Neither reads a perturbation trial from the held out animal.

In the microstimulation cohort the same machinery gives $+0.125$ against $+0.110$ for the stereotype, a difference of $+0.015$ at $p = 0.656$, which with six animals is nothing.

There is a catch in all of this, and it is the reason for the next section. A perturbation moves a population in a broadly stereotyped way, that stereotyped part dominates any score on the whole response, and a model can score well on it while knowing nothing about individual neurons. Scoring the whole response cannot tell the two apart.

4.2 Behaviour

The behavioural response transfers well. Taking the average response of the other five mice at each current and applying it to the sixth gives $\Delta R^2 = +0.572$ (95% CI $[+0.451, +0.692]$), positive in 6/6 animals, $p = 0.031$. A smooth curve fitted through current and time does about the same, at $+0.591$, and our dynamical model does about the same again, at $+0.544$. Figure 4 shows the measured and predicted curves for every held out mouse.

Two things are worth saying plainly about this. The effect is large and it holds in every animal, which is the strongest form the evidence can take with six of them. And the simplest method we could think of already achieves it. Averaging five mice predicts the sixth about as well as anything else we tried, and adding the new mouse's resting activity to the model made the prediction worse rather than better ($+0.420$). The behavioural consequence of stimulating this part of cortex is conserved enough across individuals that it does not need a model of that individual.

4.3 Population activity

The population response carries the same shape from animal to animal. The correlation between the predicted and measured time course is $+0.710$ for the group average and $+0.669$ for the model, so the timing, the rise and fall, and the dependence on current all come across. The size of the effect is another matter. At the animal level the score is $+0.256$ $[-0.105, +0.580]$ for the group average and $+0.336$ $[+0.067, +0.579]$ for our best model, with 5/6 and 5/6 animals above zero and p values of 0.156 and 0.094. With six animals that is not enough to claim that population level effect sizes transfer reliably. The shape does, and the size is unresolved.

4.4 The same is true of every model we tried

On every readout we compared each model against the best of the simple alternatives, using a paired test over animals, and nowhere does one win. For behaviour the difference is -0.047 against a smooth curve in current and time, at $p = 0.406$, and for the neural readouts the differences are small and inconsistent in sign. This holds for a smooth function of the physical variables, a manifold alignment style group average, a model built from each animal's own spontaneous dynamics, and the network. On single neurons in the microstimulation cohort the score is $+0.103$ for the average of the other mice (5/6 animals above zero, $p = 0.156$) and $+0.035$ for the dynamical model (3/6 animals, $p = 0.656$), against a ceiling of 0.91 for the same quantity.

The conserved part of the response is large and the models all find it. Nothing about that tells us whether any of them knows anything about an individual neuron, which is the next section.

5 The part that is individual

A score on the whole response is a blunt instrument, and it is blunt in a way that flatters everyone. A perturbation moves a population in a broadly stereotyped way, so a model can score respectably while handing every neuron the same answer. To ask whether anything genuinely individual carries across animals, split the measured effect into the part the recorded population shares and the part specific to one neuron:

$$\Delta_n(t) \;=\; \overline{\Delta}(t) \;+\; \delta_n(t), \qquad \overline{\Delta}(t) = \frac{1}{N}\sum_m \Delta_m(t),$$

and score models on δ alone. A stereotype borrowed from other animals gets exactly zero here, not approximately zero, because its prediction does not vary across neurons and so has no δ at all. Anything above zero is prediction of individual structure in an animal whose perturbation trials were never read.

Alongside it we compute how much of δ a recording can actually resolve. The effect is a difference of two trial means, so its noise variance is $v(1/n + 1/n_0)$ with v the trial to trial variance, n perturbed trials and n_0 control trials, all of which we know. That gives the score a perfect predictor of the true δ would reach.

One more thing about the statistics. A per animal score divides by that animal's own effect energy, so an animal whose effect is barely there can return a wildly negative number and drag a mean around with it. The headline test is therefore an exact sign test over animals, which asks whether the operator helps in an animal more often than not and cannot be moved by one extreme value. The mean and its interval are reported next to it.

Under light, individuality transfers. A shared operator acting on each neuron's own control activity is above zero in 15 of 20 mice ($p = 0.041$), with a median of $+0.026$ and a mean of $+0.033$ $[-0.034, +0.092]$, against a measurable maximum of $+0.574$. So about 6% of what is individual about a neuron's response to an intervention it has never received can be read off from what that neuron does when nothing is happening, using coefficients fitted entirely in other animals. It is a small fraction of what is there, and it is not zero, and the null control says the procedure returns -0.002 when there is nothing to find.

The network recovers more of it on average, $+0.067$ against a ceiling of $+0.574$, which is 12% rather than 6%. It is less consistent from animal to animal, 13 of 20 rather than 15, so the smaller and steadier number is the one we would defend.

Under current, it does not. The same analysis on the microstimulation cohort puts 3 of 6 animals above zero ($p = 1.000$), with a mean of $+0.002$. This is not a measurement problem. The measurable maximum there is $+0.895$, higher than in the light cohort, so the individual part is sitting there clearly resolved and no rule we fit reaches it.

That contrast is the result. The same decomposition, the same operator, the same protocol, applied to two ways of pushing on a cortex, and individuality transfers in one and not the other.

How much transfers is set by how well the animal was measured. The size of the effect under light varies enormously between animals, from essentially nothing to a third of everything measurable, and that variation is not random. It tracks the ceiling, which is fixed by trial counts and firing rates before any model is fitted, at a rank correlation of $--$ ($p = --$ over 39 animals). Animals whose individual response cannot be resolved contribute nothing, and animals where it can contribute a lot.

So split the animals in half by that ceiling, which is decided before any model is fitted and has nothing to do with how well anything scored. In the better measured half the learned operator recovers $+0.167$ of the individual part, above zero in 9 of 10 animals ($p = 0.021$), which is 22% of everything measurable there. In the worse measured half it recovers -0.033 in 4 of 10. The shared linear operator behaves the same way, $+0.065$ against $+0.002$.

A fifth of what is individual about a neuron's response to an intervention it has never received, read off from what that neuron does when nothing is happening, using coefficients fitted entirely in other animals. That is the size of the effect where the effect can be seen at all.

Two things follow from that. Averaging a per animal score over the whole cohort understates what the operator does, because the score is a ratio and animals with almost no measurable effect divide by almost nothing; pooling the errors instead, which is the usual way to combine and weights each animal by how much effect it actually has, gives $--$ for the shared operator against $--$ for the null. And the modest average says more about the recordings available today than about brains: the same analysis on recordings with hundreds of simultaneous neurons should recover a larger share.

6 What the shared rule is

The operator transfers, so something about a neuron's ordinary activity must say how a perturbation will move it, and that something must hold in every animal. It is worth asking what it is, in a form that fits in a sentence and that does not depend on any model we fitted.

So we fitted nothing. Inside each recording we correlated the individual part of each neuron's measured response with three properties read off its control trials: how fast it fires, how much its firing grows across the delay, and how strongly it distinguishes the two trial types. Each animal contributes one number and the animals are combined with an exact sign test. The control trials supplying the properties are disjoint from the ones defining the response, and the whole thing is repeated with the perturbed trials replaced by control trials as a null.

Two of the three carry, and one does not (Fig. 8).

property of the neuron	correlation	animals negative	p	null
how fast it fires	-0.221	28/38	0.005	+0.000, 17/38
how choice selective it is	-0.171	32/38	2.4e-05	-0.001, 18/38
how much it ramps	-0.003	19/38	1.000	-0.041, 25/38

The light takes most from the cells that have most: the fastest firing ones, and the ones that most sharply distinguish the two choices. The ramp on its own carries nothing, which is a useful negative control inside the same analysis, since an artefact of measurement would not know the difference between a ramp and a selectivity.

The cohort is three separate releases of the same experiment, collected years apart, and the selectivity result replicates independently in each half of it: 17 of the first release's 19 animals ($p = 7.3e-04$) and 15 of the later releases' 19 ($p = 0.019$). Firing rate replicates in the first (15/19, $p = 0.019$) and is a trend in the second (13/19, $p = 0.167$). The ramp is null in both.

Now put the microstimulation cohort through the same analysis. There, how fast a neuron fires says nothing about how much the current moves it: -0.015, negative in 4 of 6 animals, $p = 0.688$. That cohort has no two-alternative choice, so selectivity is not defined for it and only the rate is comparable.

This is the whole paper in one comparison. A shared operator can only carry individuality if a neuron's ordinary activity reports how much a perturbation will move it. Under light it does, by a rule that holds in 32 of 38 animals. Under current it does not, which is what our simulated cortex said would have to be true for the transfer to fail there, and we can now check it directly instead of inferring it.

7 From a predicted effect on neurons to a predicted change of mind

Silencing frontal cortex during the delay changes what the animal does next. If the operator has really captured the causal effect of the light, then the behavioural consequence should follow from the neural one, in an animal whose perturbation trials were never read. That is a chain we can build explicitly and then test.

For a held out animal the operator predicts, neuron by neuron, how the light will move activity. We project that predicted movement onto the animal's own choice axis, which is measured from control trials as the difference between the two trial types, and read off how far the light is predicted to push the population toward one choice. A single shared coefficient, fitted on the other animals, turns that displacement into a predicted change in the probability of licking right. Nothing in the chain reads a perturbation trial from the held out animal.

The stereotype, meaning the average behavioural change of the other animals at the same dose and side, already does well: a median $+0.511$ across animals, positive in 13 of 20. Adding the neural chain changes that by -0.006 at $p = 0.656$, which is nothing.

We expected the chain to help and it does not, and the reason is the same one the rest of the paper is about. Behaviour is a population level readout. What the light does to a population is stereotyped enough across animals that the other animals already predict it, so the animal's own neurons have nothing left to add. Individuality lives below that level, in which neuron is moved and by how much, and a behavioural readout integrates precisely that away. A model can therefore predict an animal's change of mind without knowing anything about that animal, which is worth stating clearly for anyone who would take behavioural transfer as evidence that a model has captured an individual.

One caveat about which animals this is measured in. The second and third releases of the cohort use brief unilateral inhibition of the kind their authors introduced to show that premotor dynamics recover from it [11], and in those recordings the light barely moves behaviour at all, so there is no behavioural effect to predict. The numbers above are from the first release, where silencing costs the animal 5.6 percentage points in the fraction of correct trials.

8 How many animals an operator needs

If what carries between animals really is one operator, fitting it on more animals should make it better on a new one. We fitted the shared operator on random subsets of the training animals, at every size from one upwards, and scored it on the individual part of the held out animal's response.

The curve is orderly and it starts badly. One animal gives -25.966, which is not merely useless but far worse than predicting nothing at all, because a single animal's idiosyncrasies get mistaken for a rule and then applied confidently to somebody else. Five animals are still deeply negative, and the average across animals only reaches zero at 38. Across the range it tracks the logarithm of the cohort size at $r = +0.77$ and has not flattened.

That average is a mean of ratios and one animal with almost no measurable effect can hold it down, so the number to read is how many animals the operator helps in. At 38 animals it is above zero in 29 of 382 mice ($p = 0.003$); at five animals it is above zero in fewer than half.

This is the clearest practical statement in the paper. A study with five animals would not have found this effect and would have reported, with a straight face, that nothing transfers. The returns are still being collected at 38 animals, which is an argument for cohorts over deeper recordings in a handful of animals.

The same holds for the trained network, which cannot be refitted for every subset but can be compared between two runs that differ only in how many animals it had to learn from. Holding out the same – animals and scoring them identically, the network trained on – other animals scores $--$ against $--$ for the one trained on –, better in – of – of them ($p = --$).

9 Where the failure comes from

Knowing that single neurons fail is less useful than knowing what is missing, so we took the shared rule apart. Everything in this section is the microstimulation cohort, where the failure is.

One number per neuron. We froze the shared rule at the value fitted on the other animals and changed nothing about it. Then we allowed the held out animal one scalar per neuron, fitted on that animal's stimulation trials. This is not a prediction, because it uses data the protocol forbids. It is a way of asking what the shared rule already knows. A scalar per neuron cannot invent structure in time, and it cannot invent structure across currents, because each neuron's predicted time course and dose dependence are fixed by the shared rule.

That one number per neuron takes the score from $+0.035$ to $+0.422$ $[+0.355, +0.499]$, positive in 6/6 animals, with an animal level difference of $+0.387$ at $p = 0.031$, which is the floor for six animals. Allowing an offset as well gives $+0.511$. So the shared rule was already getting the shape of each neuron's response right. It was getting each neuron's amplitude wrong.

That number is not guessable. Having found the missing quantity, we tried to predict it without touching any stimulation trial in the held out animal. The predictors were everything non interventional we had, including each neuron's depth and its offset from each stimulating contact, its firing statistics, and its spontaneous coupling to the neurons near each contact, which is the natural linear response proxy for how strongly the stimulus should reach it. Shrinkage was chosen by a nested leave one animal out loop inside the training animals.

Across all 949 neurons, predicted and required amplitudes correlate at only $r = +0.174$, and applying the prediction moves the score from $+0.035$ to $+0.065$, which does not survive an animal level test ($p = 0.562$). An exact per neuron amplitude would give $+0.422$. So the gap is real, it is exactly one number per cell wide, and nothing we can observe without stimulating fills it.

Why it is not guessable. The obvious guess is distance. A neuron near the electrode should respond more. Pooled over every session and every condition, that is not what the data show. The correlation between how far a neuron sits from the stimulating contact and how much it responds is -0.013, with no peak at the electrode and no falloff, just a broad elevation spread over more than a millimetre in both directions (Fig. 2b). This is the signature of low current stimulation driving axons of passage, which reaches a sparse and scattered set of cells rather than a dense local sphere, as direct imaging of stimulation evoked activity also reports [9]. Which cells those are is a property of where that particular electrode ended up in that particular brain. There is no species level rule to learn at that resolution.

10 A simulated cortex that tests the explanation

The explanation above is testable, so we built a cortex where we control the thing we think matters. Neurons sit along a depth axis and are connected by a rule shared across animals, plus a private part. Stimulation at a contact injects current, and we can choose who receives it.

In the *local* setting the electrode drives neurons by a smooth rule of distance, so which cells respond is the same rule in every animal. In the *sparse* setting the electrode drives a scattered subset drawn fresh for each animal and each contact, so the amount of drive is still shared while the identity of the recipients is private to that implant. We then vary how many neurons are recorded at once and score transfer exactly as we do for the mice.

The two settings come apart (Fig. 5). With a shared local rule the score averages $+0.679$ across population sizes, every animal is above zero at every size, and the score climbs steadily as more neurons are recorded, with a correlation of $+0.987$ between the score and the log of the population size. With private recruitment the average falls to $+0.409$, the trend with population size becomes erratic ($+0.582$), and the spread across animals roughly triples, from a mean interval width of 0.15 to 0.52. At the population sizes found in our mice the private setting is at its most unstable, with intervals that comfortably span zero.

The reading is not the one we expected. Private recruitment does clearly damage single neuron transfer, costing about $+0.270$ in score at matched population size and making the result far less reliable from animal to animal. But it does not by itself abolish transfer in this simulation, so it is not a complete account of the failure in the mice, which is more severe. The honest reading is that private recruitment is one contributing cause among others, and that small simultaneously recorded populations are another.

The clean trend in the shared setting is the useful part for planning. When the rule governing who gets driven is shared, recording more neurons at once helps in a very regular way. Our mice have between 8 and 33 neurons per session, at the bottom of the range we simulated.

10.1 Two things a perturbation can share

Running the decomposition of Section 5 on the same simulated cortex says something the score on the whole response could not. There are two separate ways a perturbation can carry individual structure between animals, and they can be switched on and off one at a time.

The first is where the perturbation lands. Turning the recruitment from a smooth rule of position into a scattered set redrawn for each animal takes the recovered fraction of the individual part from 46% to 27%.

The second is how strongly a cell reacts given what it was already doing, and it is the one that surprised us. In the simulation a cell's ongoing firing rate reports how strongly the network drives it, so it also reports how much any perturbation will move it, and that is a rule every animal obeys. It survives private recruitment. We tried to switch it off by making every simulated cell equally responsive, and the recovered fraction went up rather than down, because removing that variation makes position a cleaner cue. There is no setting of this simulator in which private recruitment alone drives transfer of the individual part to zero.

The microstimulation cohort sits below the worst case the simulator can produce. So private recruitment is part of the story there and not all of it, and the missing piece has to be the second mechanism: for electrical stimulation, a neuron's ongoing activity must fail to say how much the stimulus will move it, while under light it must succeed.

That was a prediction when we wrote it, and Section 6 is the test. It holds. Under light a neuron's firing rate predicts how much it is suppressed in 28 of 38 animals, and how sharply it distinguishes the two choices predicts it in 32 of 38. Under current the same measurement gives -0.015 in 4 of 6 animals, which is nothing. The simulator told us which quantity to go and measure, and the measurement came out the way it had to for the two cohorts to differ as they do.

One detail of our guess was wrong, and it is worth recording. We expected the light to act on the cells that were ramping hardest. It does not: the ramp on its own predicts nothing (19/38, $p = 1.000$). What it acts on is selectivity and rate.

11 When can transfer work at all

It helps to know when this kind of transfer is possible in principle. Write the model as a shared latent state that every animal shares, a shared rule for how it evolves, a shared rule for how a stimulus pushes it, and an animal specific map from that state to the neurons we happen to record:

$$z_{t+1} = z_t + \Delta t \left[F(z_t, u_t) + G(z_t)\, a_t + R_i(z_t) \right], \tag{1}$$

$$y_t^{(i)} \sim \text{Poisson}\big(\text{softplus}(C_i z_t + b_i)\big), \qquad \text{behaviour}_t = D(z_t). \tag{2}$$

Proposition 1 (Transfer is pinned down up to the symmetries of the shared flow)**.** *Fix F and G. Suppose a new animal's unperturbed law is matched both by (C_i, R_i) and by (C_i', R_i'), with C_i of full row rank and the* *latent state reachable under the exogenous inputs. Then there is an invertible T with $TF(z,u) = F(Tz,u)$ for all z and u, such that $C_i' = C_i T^{-1}$ and $R_i' = TR_iT^{-1}$. The two give predictions for the same intervention that differ by $C_iT^{-1}\big(TG(Tz)a - G(z)a\big)$. If the only such T is the identity, the predicted response is unique. Otherwise it is ambiguous by exactly that group, unless G happens to be equivariant under it.*

The argument is a change of variables $z \mapsto Tz$ in the two equations above, together with the requirement that the unperturbed law be preserved; full rank of C_i and reachability are what rule out reparameterisations that act only on directions the data never visit. Two things follow, and both showed up in practice. Resting activity sits near a fixed point, where the flow is weak and close to isotropic, so a freely fitted C_i has a lot of room and the new animal's frame is poorly determined. That room can be closed by construction, by generating C_i from a shared function of per neuron features instead of fitting it freely, which is why a deliberately symmetric simulated teacher still transfers in our tests. The transfer is then only as good as those features are informative about how strongly the stimulus reaches a given cell, and in this preparation they are not.

12 How we kept ourselves honest

Several of the checks below changed a number in this paper, which is why they are worth listing.

The one that changed the most was the overlap between the control trials the model reads and the control trials that define its target. We had a positive result before we noticed it. Splitting those trials in two, so that no trial appears on both sides, cut the recovered fraction of the individual part roughly in half, and everything reported here is from the split version. The check that would have caught it earlier is the null control we now run alongside every fit: replace the perturbed trials with a second group of control trials, so the answer is known to be zero, and confirm the procedure returns zero. It does, at -0.002 under light and -0.002 under current. Before the split it did not.

The split half ceiling estimator was validated against simulations with known signal and noise across six regimes of noise level and trial count, agreeing with the attainable score to within 0.06. The ceiling for the individual part is computed a second, independent way, analytically from trial counts and firing rates, and the two agree.

Stimulation trials are delivered under a quiescence criterion, so inter trial windows sampled at random carry up to three times the pre stimulus firing rate of a real trial. Since predictions start from unperturbed initial conditions, that would have inflated every effect we report. Candidate windows are matched to stimulation trials on a joint quantile grid of pre stimulus firing rate and wheel speed and then trimmed to match on the mean, and any session still mismatched by more than a factor of 1.25 is dropped. The released audit reports 0 remaining problems.

Seeds are derived with a stable checksum instead of Python's per process hash, so two independent builds of the analysis tensors agree exactly, and the results table reproduces bit for bit across re runs.

Deleting one middle current from the training animals turns out to be close to interpolating between its neighbours, so we also delete the top of the range, the bottom of the range, a contiguous block, whole contact depths, and specific combinations of current and depth. Table 3 gives the behavioural result under each. Transfer to a genuinely unseen current is weaker than the interpolating case, and with six animals we do not claim it.

13 What this means

The similarity that people describe between animals is not only in the shape of the activity. Stimulating this part of cortex moves different mice in a way that is conserved enough to predict the behavioural consequence in an animal that has never been stimulated, and conserved enough to predict the shape of the population response. That is a statement about mechanism, and mechanism is what a perturbation tests.

The same rule stops short of the individual cell, and it stops short in a specific way. It knows when a neuron will respond and how that response grows with current. It does not know how much any particular neuron will respond, and for electrical stimulation that is because the set of neurons the electrode reaches belongs to the implant rather than to the species.

Under light it does not stop there, and the reason is now something we can name rather than fit. The light takes most from the cells that most sharply distinguish the two choices and from the cells that fire fastest, and it does that by the same rule in 32 of 38 animals. That is a statement about what a perturbation of this kind does to a cortex, not about our model, and it is the reason a shared operator reading each neuron's own ordinary activity can recover part of what is individual about that neuron's response. How much it recovers grows with the size of the cohort it was fitted on and with how well each animal was measured, and both of those are things an experiment can change.

It also says what a perturbation has to do for any of this to work. Electrical stimulation reaches cells for reasons that have nothing to do with what those cells were doing, and correspondingly nothing individual transfers. An optogenetic perturbation acts on a population defined by what it expresses and how active it is, and the effect on a named cell becomes partly predictable from that cell's own behaviour. The design of the perturbation decides whether there is a rule to share.

The practical reading is mostly about how to ask. Scoring a cross-animal model on the whole response tells you almost nothing, because the stereotyped part dominates it and a stereotype costs nothing to build, so the split used here, or something like it, is what makes the question answerable at all. Cohorts below about eight animals cannot see the effect and would report that nothing transfers. For anyone calibrating a stimulation protocol, the behavioural and population level effects seen in one animal remain a reasonable guide to the next, while the effect on any named cell is a guide only when the perturbation acts through something the animals hold in common.

14 What we could not settle

The positive result is small. A few per cent of the measurable individual part is real and repeatable across animals, and it is a long way from the whole of it. We cannot yet say how much of the remainder is genuinely private wiring and how much is beyond the operator we happened to write, and the honest position is that both contribute.

Six animals is the binding constraint on everything in the microstimulation cohort. The smallest p value available from an exact test over six animals is 0.031, so results that hold in all six are as strong as that design allows, and results that hold in four or five are suggestive at best.

Our simulated cortex never drives transfer of the individual part to zero, at any setting we could find, and the microstimulation cohort sits below its worst case. So private recruitment is part of the account there and not all of it. The piece we cannot see directly is whether a neuron's ongoing rate reports how much a perturbation will move it, which our simulator assumes and which the light cohort behaves as though it obeys. Measuring responsiveness and ongoing rate in the same cells under both kinds of perturbation would settle that, and it does not need a new dataset so much as a new analysis of one.

The ceiling on the result itself would lift with things an experimenter chooses. The cohort curve has not flattened at twenty animals. And how much transfers in an animal tracks how well that animal was measured, so recordings with hundreds of simultaneous neurons should recover a larger fraction than the 1668 neurons spread over 109 sessions that we had.

Finally, this is two cohorts, one species, two cortical areas and two ways of perturbing. The contrast we report is between those two, and a third perturbation could easily land somewhere we did not anticipate.

Data and code

Both datasets are public [12, 13]. Everything in this paper is rebuilt by `make all`, from downloading the raw files with a checksummed manifest through the leakage audit, the results table, the simulations, the figures and the numbers quoted in the text, which are generated directly from the result files.

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Table 2: What transfers to a held-out animal. Scores are averaged over animals, with a bootstrap over animals for the interval and an exact sign-flip permutation over animals for the p value. A score of zero is the model that says the stimulus does nothing. With six animals the smallest attainable p value is 0.031.

readout	method	ΔR^2	95% CI	r	animals > 0	p
behaviour	average of the other mice	+0.572	[+0.451, +0.692]	+0.652	6/6	0.031
	smooth curve in current and time	+0.591	[+0.477, +0.705]	+0.680	6/6	0.031
	shared rule (this paper)	+0.544	[+0.395, +0.674]	+0.660	6/6	0.031
	shared rule + resting activity	+0.420	[+0.204, +0.613]	+0.647	5/6	0.062
population	average of the other mice	+0.256	[-0.105, +0.580]	+0.710	5/6	0.156
	shared rule	+0.225	[-0.088, +0.494]	+0.669	4/6	0.188
	shared rule through own dynamics	+0.108	[-0.252, +0.446]	+0.668	3/6	0.562
	the same + predicted responsiveness	+0.336	[+0.067, +0.579]	+0.668	5/6	0.094
single neurons	average of the other mice	+0.103	[+0.010, +0.201]	+0.297	5/6	0.156
	shared rule	+0.093	[+0.010, +0.177]	+0.275	4/6	0.156
	shared rule through own dynamics	+0.035	[-0.078, +0.158]	+0.275	3/6	0.656
	the same + predicted responsiveness	+0.075	[-0.051, +0.192]	+0.275	5/6	0.281

Table 3: Behavioural transfer when the stimulus setting is also removed from every training animal. Deleting a middle current is close to interpolating between its neighbours; the other rows are harder.

held out from all training animals	ΔR^2	95% CI	animals > 0	p
nothing	+0.544	[+0.395, +0.674]	6/6	0.031
one middle current	-1.761	[-5.260, +0.283]	2/4	0.625
a block of three currents	+0.097	[-0.631, +0.658]	4/6	0.812
the top of the current range	+0.289	[-0.674, +0.788]	2/3	0.500
the bottom of the current range	-15.690	[-24.463, -6.756]	1/6	0.062
all superficial contacts	-55.259	[-68.460, -37.280]	0/5	0.062
all deep contacts	-14.756	[-24.174, -5.837]	1/5	0.125
high current at superficial contacts	-0.435	[-1.998, +0.635]	4/6	0.906

Figures

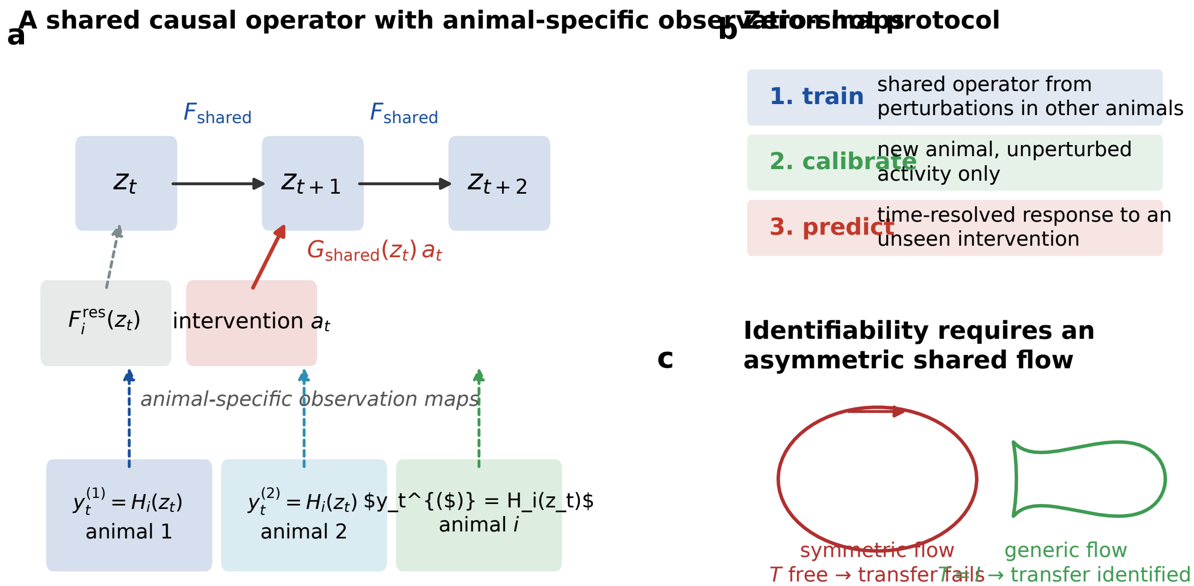

Figure 1: The question and the protocol. **a**, The model separates a rule shared by all animals, covering both how the latent state evolves and how a stimulus pushes it, from the animal specific map onto the neurons that happen to be recorded. **b**, The protocol. The shared rule is learned from stimulation in other animals, the new animal contributes only resting activity, and the response to a stimulus it has never received is then predicted. **c**, Proposition 1. A symmetric shared flow leaves a reparameterisation free and the prediction is ambiguous, while a generic flow pins the observation map down.

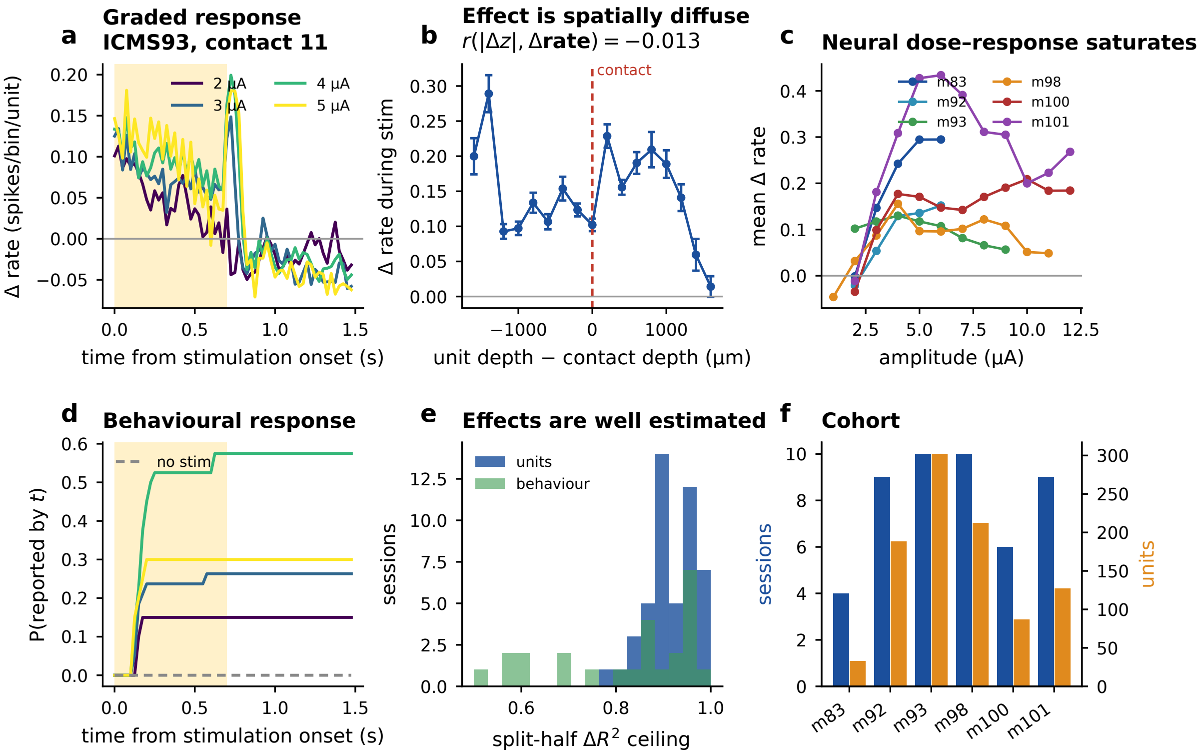

Figure 2: Microstimulation in mouse somatosensory cortex. **a**, The population response grows with current. **b**, Pooled over every session and condition, how much a neuron responds is essentially unrelated to how far it sits from the stimulating contact ($r = -0.013$). The response is spread out rather than centred on the electrode. **c**, Within animal dose response, which rises and then saturates. **d**, The behavioural response, meaning the probability of having reported the stimulus by time t. **e**, Split half ceilings for every session. **f**, Composition of the cohort.

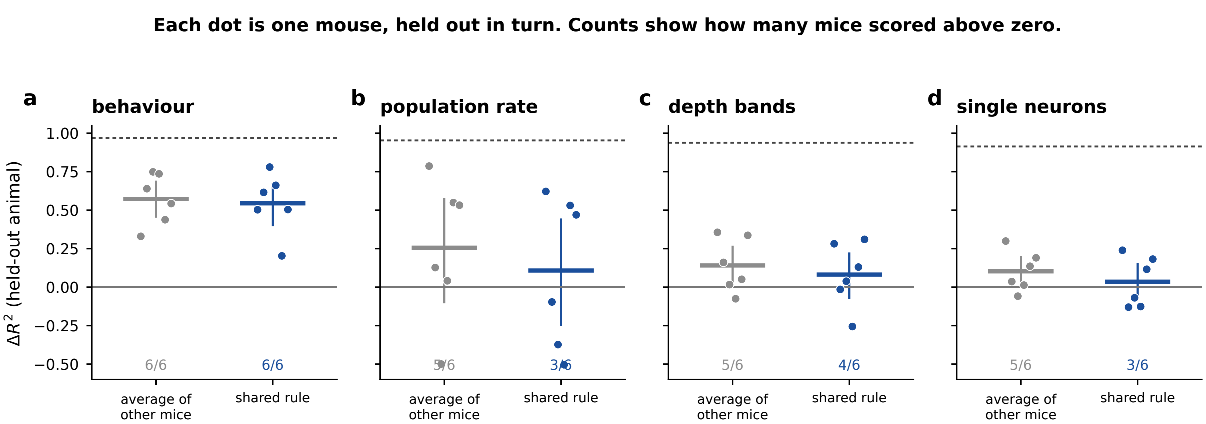

Figure 3: What transfers, by readout. Each point is one animal, held out in turn. Behaviour transfers in every animal. Population activity carries the right shape with unreliable size. Single neurons do not transfer. Dashed lines mark the highest score the trial counts allow.

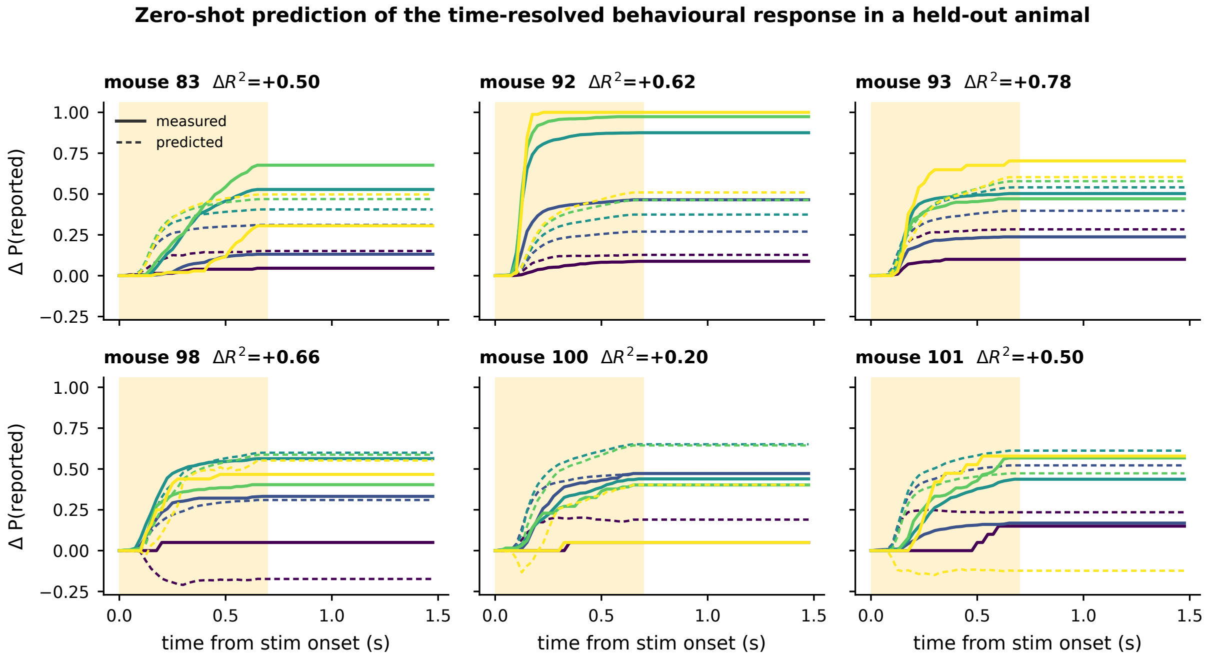

Figure 4: Predicting behaviour in a mouse the model never saw stimulated. Measured (solid) and predicted (dashed) change in the probability of having reported the stimulus, coloured by current. The model saw no stimulation trial from the animal shown in each panel.

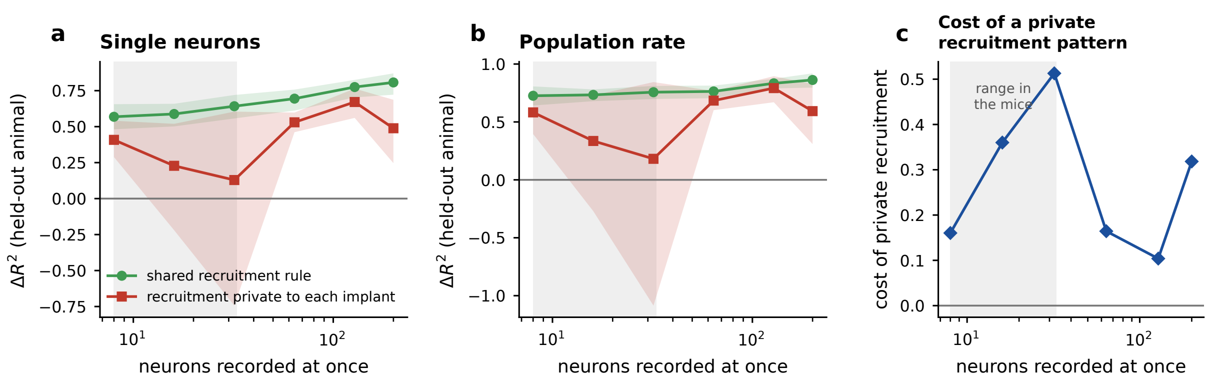

Figure 5: A simulated cortex tests the explanation. **a**, When the electrode drives neurons by a rule shared across animals, single neuron transfer works at every population size and improves as more neurons are recorded. When it drives a scattered set private to each implant, transfer fails at the population sizes found in the mice and recovers once enough neurons are recorded. **b**, The same comparison for the population readout, which is much less sensitive to how recruitment is arranged. **c**, The shaded band marks the range of simultaneously recorded neurons in the real dataset.

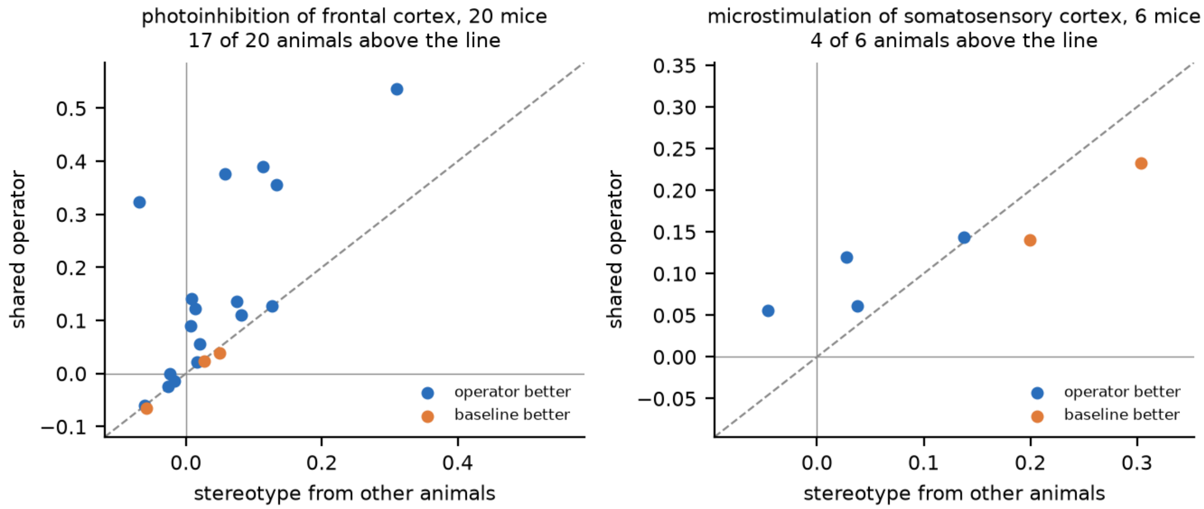

Figure 6: Predicting the whole response of individual neurons in an animal never seen perturbed. Each point is one animal, held out in turn. The horizontal axis is the average of the other animals at the same setting, which is the strongest simple thing to do, and the vertical axis is the shared operator. Points above the dashed line are animals where the operator wins. Left, light in frontal cortex. Right, current in somatosensory cortex.

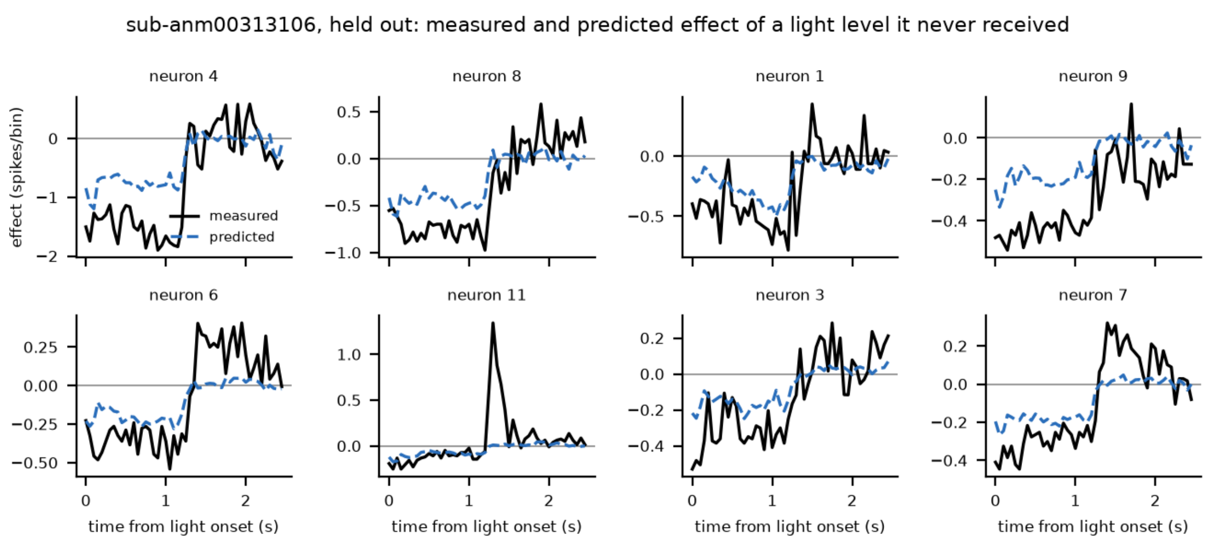

Figure 7: What a prediction looks like. Eight neurons from one held out mouse, under a light level that mouse never received during training, ordered by how strongly the light moved them. Black is measured, blue is predicted. The model was trained on other mice and saw only this mouse's control trials. It gets the sign, the time course including the release when the light goes off, and the ordering across neurons. It under-predicts how far each neuron moves, which is the paper in one picture: the rule is shared and the magnitudes are not.

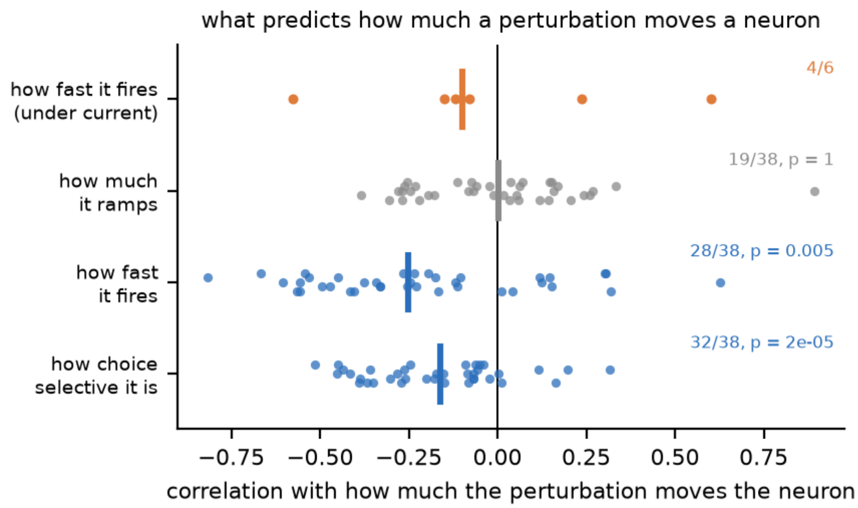

Figure 8: The shared rule, measured rather than fitted. Each dot is one animal: the correlation, computed inside that animal's recordings, between the individual part of a neuron's response and a property read off its control trials. Thick bars are medians. The light suppresses the cells that most sharply distinguish the two choices and the cells that fire fastest, in nearly every animal. How much a cell ramps predicts nothing, which is a negative control inside the same analysis. Under microstimulation (orange) there is no such relationship, which is why nothing individual transfers there.

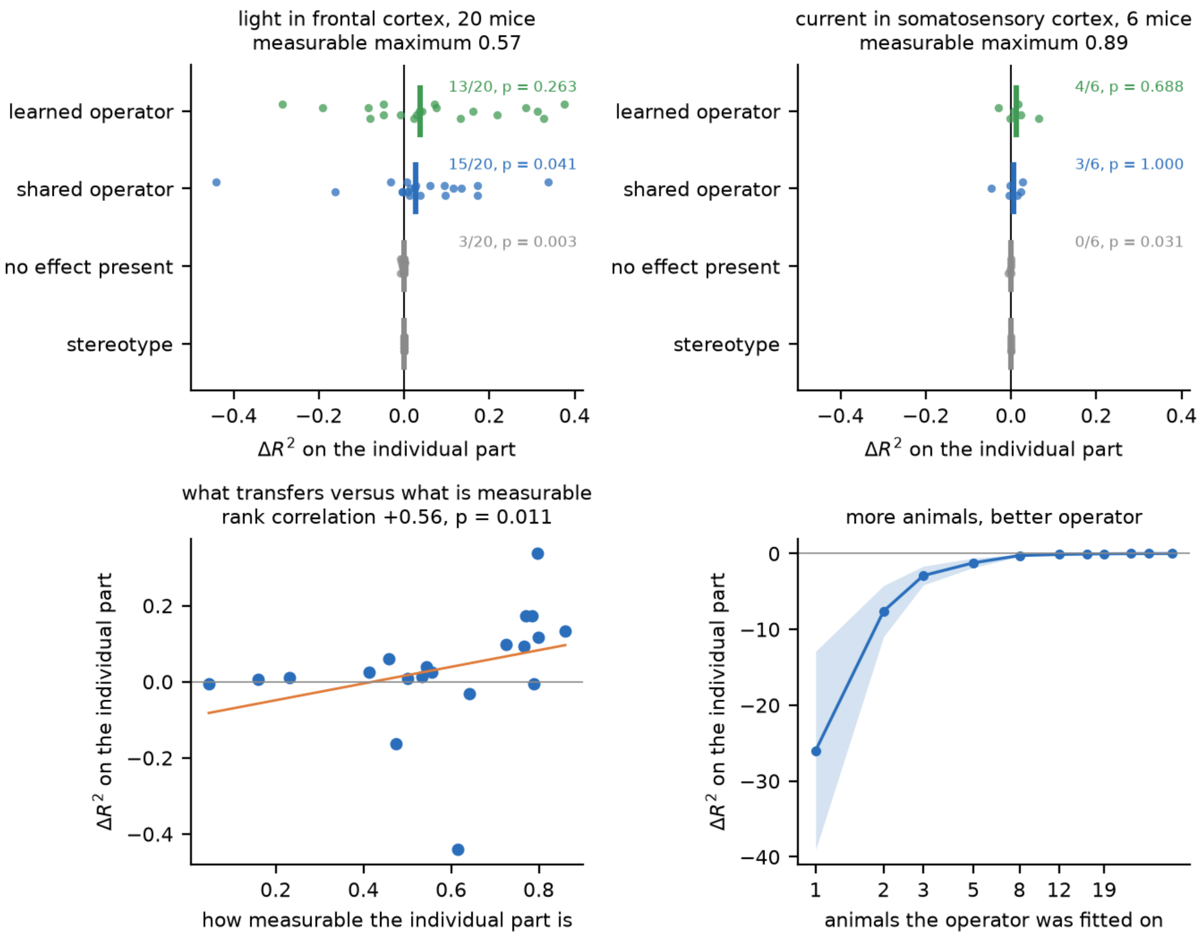

Figure 9: The part of the response that belongs to one neuron. Every model is scored on δ, what is left of the effect after removing what the whole recorded population does together. A stereotype borrowed from other animals is at exactly zero there by construction. **a**, Light in frontal cortex. Each dot is one mouse, the thick bar is the median, and the row marked “no effect present” is the identical analysis with the light trials replaced by a second group of control trials, so the answer is known to be zero. **b**, Current in somatosensory cortex, on the same axis. The measurable maximum is higher here and nothing transfers. **c**, How much transfers in a mouse against how well that mouse’s individual response could be measured at all, a quantity fixed by trial counts and firing rates before any model is fitted. **d**, The operator fitted on random subsets of the training animals. One animal is much worse than predicting nothing; the curve crosses zero near eight and has not flattened at nineteen. Shading is the standard error over held out animals.

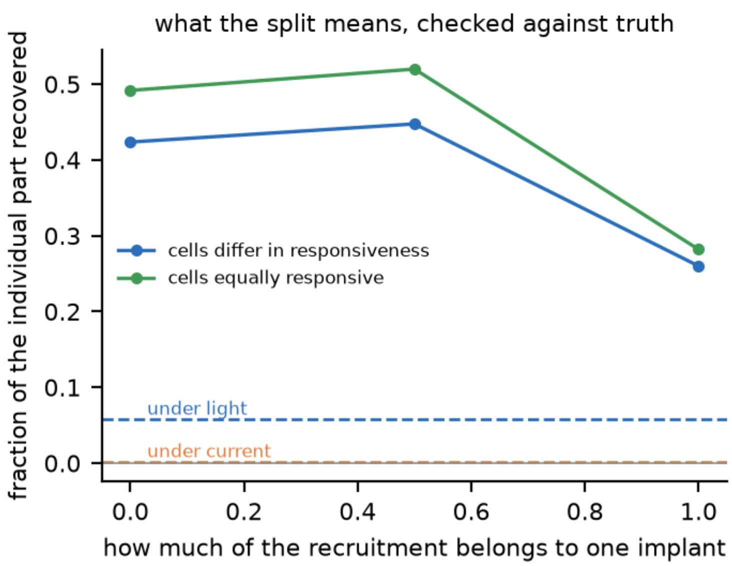

Figure 10: What the split means, checked against a cortex where we know the answer. Turning the recruitment from a smooth rule of position into a scattered set redrawn for each animal costs about 46 minus 27 per cent of the recoverable individual part, and making every simulated cell equally responsive does not help, so private recruitment alone never drives the simulation to zero. Dashed lines are the two real cohorts, both of which sit below anything the simulator produces.